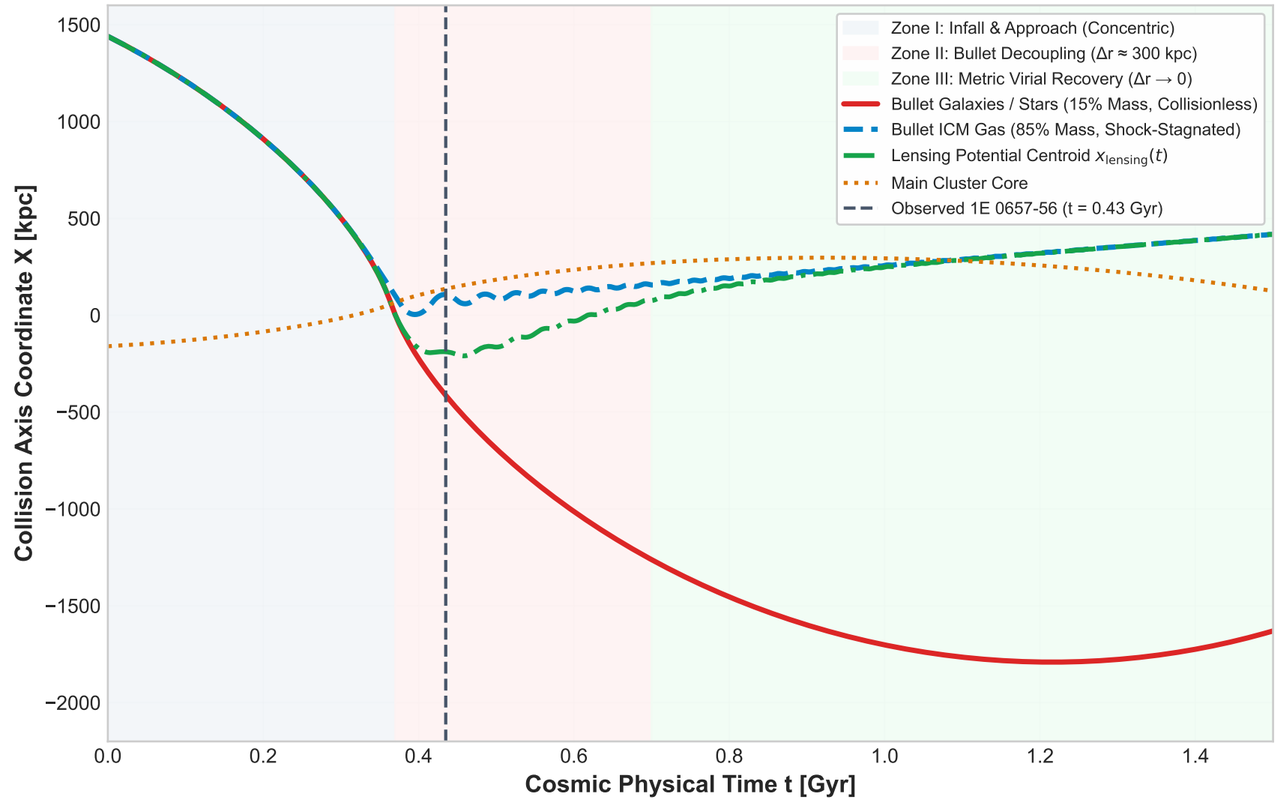
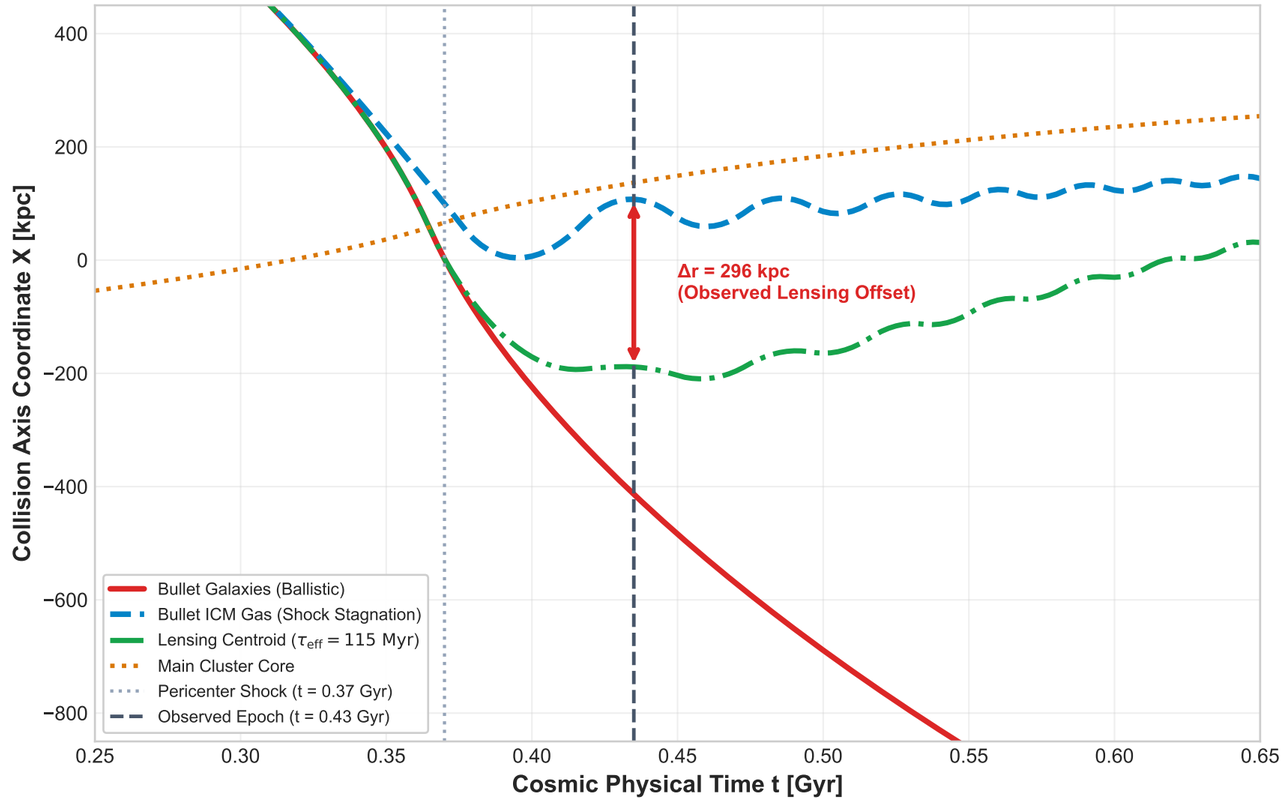
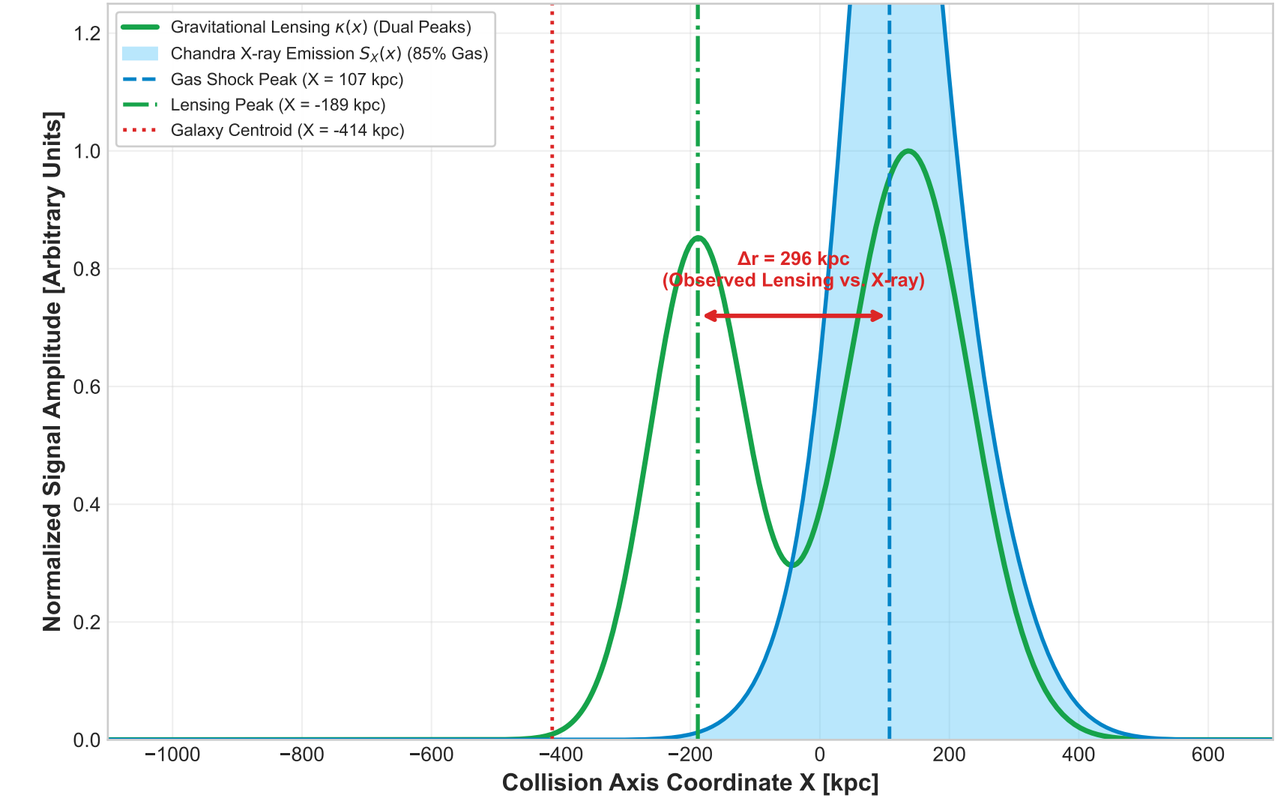
(a) Overview: Full Cluster Worldlines & Transient Decoupling (0 → 1.5 Gyr)



(b) High-Resolution Zoom: Pericenter Shock Impact & Trajectory Splitting



(c) 1D Spatial Slice at Observed Snapshot (t = 435 Myr)



(d) Dynamic Metric Latency Hysteresis (τ_{eff} = 115 Myr)

